

CSE350: Digital Electronics & Pulse Techniques

Lecture 09: ADC Intro, Flash ADC

Analog: Information or data that can be any real value at any instant of time. Physical world signals are of analog nature.



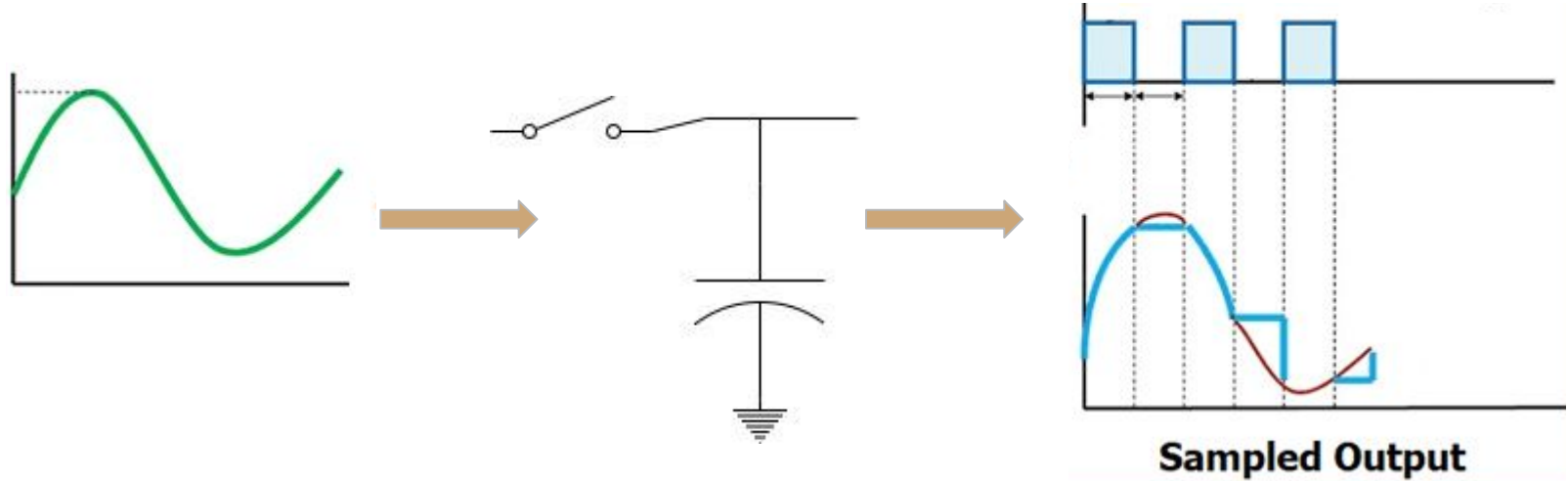
Digital: Information in the form of binary bit streams. Computers & all digital devices need digital signals to operate.



ADC is the process of converting analog signals to digital signals. It can be summarized as below:



Sample & Hold: The analog signal is sampled at regular intervals of a clock pulse. A capacitor is used to hold the peak value of the preceding interval until next one arrives. Thus, instead of all the real values of the original signal, we get a signal that has reduced specific values, i.e. a discrete signal.



Quantization: A process to assign some fixed values to the discrete analog signal from sample & hold process. These fixed values are called levels of quantization. Depending on the number of bits to be used in representing the digital signal, the level number varies.

If we use n bits, then number of **quantization levels would be** 2^n

The difference between any two consecutive levels are equal and called the **resolution of quantization**. This can be expressed as,

$$\Delta = \frac{V_{max} - V_{min}}{2^n}$$

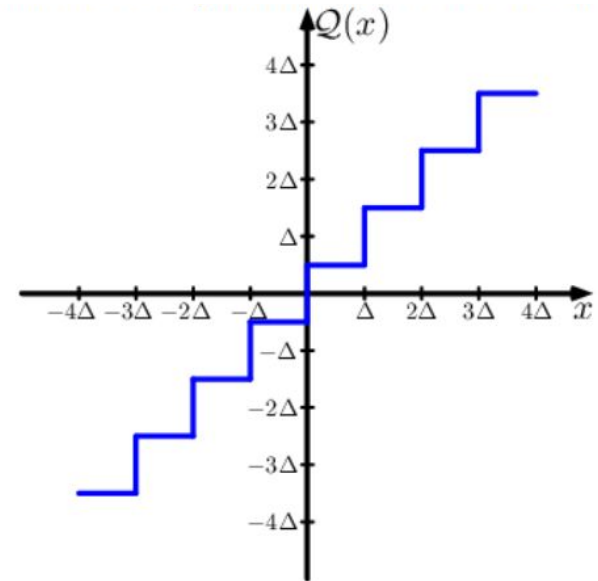
Where, V_{max} & V_{min} are the maximum and minimum values of the analog signal respectively.

Depending on the method used to assign analog values to these levels, quantization can be of the following two types:

1. **Mid-rise Quantization**
2. **Mid-tread Quantization**

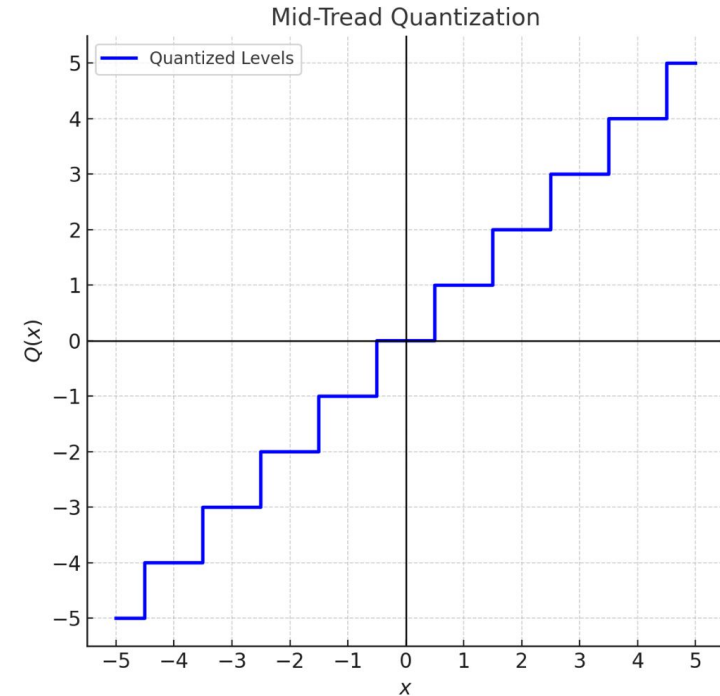
Mid-rise Quantization: A type of quantization scheme used in digital signal processing, where the range of the input signal is divided into a set of equal-sized intervals, and the output levels are assigned to the midpoints of these intervals. The quantization levels are symmetric about zero but **do not include a quantization level at zero (midpoint between minimum & maximum)**. For a signal quantized into 8 levels with a range from -4 to 4:

- Quantization intervals: $[-4, -3), [-3, -2), \dots, [3, 4)$
- Midpoints (output levels): $-3.5, -2.5, -1.5, -0.5, 0.5, 1.5, 2.5, 3.5$



Mid-tread Quantization: A type of quantization scheme used in signal processing, where the range of the input signal is divided into intervals, and the output levels are assigned to the midpoints of these intervals, **with one of the levels located exactly at zero (midpoint between minimum & maximum)**. For a signal quantized into 8 levels with a range from -4 to 4:

- Quantization intervals: $[-4.5, -3.5), [-3.5, -2.5), \dots, [2.5, 4)$
- Midpoints (output levels): -4, -3, -2, -1, 0, 1, 2, 3



Encoding: This is the process of assigning binary bit streams to the quantization levels.

For example, the mid-rise quantization levels as showed in previous slide, will be assigned the following 3 bit values:

-3.5: 000 (0)

-2.5: 001 (1)

-1.5: 010 (2)

-0.5: 011 (3)

0.5: 100 (4)

1.5: 101 (5)

2.5: 110 (6)

3.5: 111 (7)

Flash ADC: A very fast converter that requires only one clock cycle to convert the analog data to digital.

We know, for comparators,

$V_+ > V_-$ then output is high

$V_+ < V_-$ then output is low

$$V_{ref}^+ = 3$$

$$V_{ref}^- = -5$$

$$I = \frac{V_{ref}^+ - V_{ref}^-}{8R} \rightarrow IR = \frac{3 - (-5)}{8} = 1V$$

$$V_0 - V_{ref}^- = IR \rightarrow V_0 = V_{ref}^- + IR = -5 + 1 = -4V$$

$$V_1 = V_0 + IR = -4 + 1 = -3V$$

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$$V_7 = V_6 + IR = V_{ref}^+ = 2 + 1 = 3V$$

